

CSCI 361: Exercise # 8

Your Name: _____

Due: Thursday, Oct 16, in class

Assigned reading for this lecture:

- Sipser Third Edition Chap 3.1 (Turing Machines)

Questions:

1. Consider the definition of the transition function of a Turing machine. What does $\delta(q, a) = (r, b, L)$ mean where q, r are the states of a Turing machine and $a, b \in \Sigma$.
2. What is the difference between a Turing decidable and Turing recognizable language?
3. Consider the TM in Figure 3.10 (Example 3.9). State the sequence of configurations it enters on the input 01#01.